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Studierichting: Informatica
Oefeningenreeks 6B nr. 12

$$s_k = \frac{x_1 + x_k}{2} \cdot k$$

$$s_{k+1} = \frac{x_1 + x_{k+1}}{2} \cdot (k+1)$$

$$s_{k+2} = \frac{x_1 + x_{k+2}}{2} \cdot (k+2)$$

$$s_k - 2 \cdot s_{k+1} + s_{k+2} = \frac{k \cdot (x_1 + x_k)}{2} - \frac{2 \cdot (k+1)(x_1 + x_{k+1})}{2} + \frac{(k+2) \cdot (x_1 + x_{k+2})}{2} =$$

$$= \frac{kx_1 + kx_k - (2k+2)(x_1 + x_{k+1}) + kx_1 + kx_{k+2} + 2x_1 + 2x_{k+2}}{2} =$$

$$= \frac{kx_1 + kx_k - (2kx_1 + 2kx_{k+1} + 2x_1 + 2x_{k+1}) + kx_1 + kx_{k+2} + 2x_1 + 2x_{k+2}}{2} =$$

$$= \frac{2kx_1 + kx_2 - 2kx_1 - 2kx_{k+1} - 2x_1 - 2x_{k+1} + kx_{k+2} + 2x_1 + 2x_{k+2}}{2} =$$

$$= \frac{kx_k - 2kx_{k+1} - 2x_{k+1} + kx_{k+2} + 2x_{k+2}}{2} =$$

$$= \frac{kx_k - 2x_{k+1}(k+1) + x_{k+2}(k+2)}{2}$$

Als:

$$x_k = x_{k-1} + v$$

$$x_{k+1} = x_k + v$$

$$x_{k+2} = x_k + 2v$$

Dan:

$$\frac{kx_k - 2x_{k+1}(k+1) + x_{k+2}(k+2)}{2} =$$

$$= \frac{k(x_{k-1} + v) - 2(k+1)(x_k + v) + (k+2)(x_k + 2v)}{2}$$

$$= \frac{kx_{k-1} + kv - 2(kx_k + kv + x_k + v) + (kx_k + 2kv + 2x_k + 4v)}{2} =$$

$$= \frac{kx_{k-1} + kv - 2kx_k - 2kv - 2x_k - 2v + kx_k + 2kv + 2x_k + 4v}{2} =$$

$$= \frac{kx_{k-1} - kx_k + kv + 2v}{2} =$$

Als $x_k = x_{k-1} + v$, dan:

$$= \frac{kx_{k-1} - k(x_{k-1} + v) + 2v + kv}{2} =$$

$$= \frac{kx_{k-1} - kx_{k-1} - kv + kv + 2v}{2} =$$

$$= \frac{2v}{2} = v$$

References